



JavaScript Origins

Brief Overview

This note covers **JavaScript** and was created from an [uploaded document](#). It covers early web browsers, the birth of JavaScript, ECMAScript standardization, scripting basics, and real-world impact.

Key Points

- **JavaScript** emerged from Mosaic & Netscape's push for client-side interactivity.
 - Brendan Eich's 10-day sprint produced LiveScript → **ECMAScript** and the first language spec.
 - JavaScript vs. Java: interpreted vs. compiled, browser vs. server usage, and marketing origins.
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Early Web Browsing: Mosaic & Netscape (1995)

Context

- 1995: pre-Tupac, early Windows 95 era.
- HTML pages were **static** – pure *form* with **no** built-in functionality.
- Any calculation, data lookup, or unit conversion required a **round-trip** to a back-end server:
 1. Browser sent a request to the server.
 2. Server performed the computation (business logic).
 3. Server returned a new HTML page with the result.

Vision of Netscape

- Desired a **dynamic web** with animations and real-time interaction.
 - Needed to **move logic from server to client**, i.e., run code **inside the browser**.
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Birth of JavaScript

Motivation & Requirements

- Create a **small scripting language** that:
 - Is **simple** and **easy to learn**.
 - Allows **non-programmers** to add interactivity to web pages.

Brendan Eich's 10-Day Sprint

- Brendan Eich was hired by Netscape.
- He **produced the first version of JavaScript in just 10 days**.

Naming Evolution

Original name	Microsoft's counterpart	Standardized name
LiveScript	JScript	ECMAScript

- The term **ECMAScript** is the official specification; versions are referenced as **ES6**, **ES7**, etc.

ECMAScript Standardization

ECMAScript – The formal standard that defines the core language features of JavaScript.

- Different browser vendors initially implemented slightly divergent variants (e.g., JScript).
- European developers pushed for a **single, unified spec**, leading to the ECMAScript standard.

Version Naming

- Community usually refers to language releases as **ES5**, **ES6**, **ES7**, ... rather than **JS5**, **JS6**.
- “ES” comes directly from **ECMAScript**.

Scripting Analogy

A **script** tells the **actors** (HTML elements) what to do, just like a play script directs performers on stage.

Play-style Example

1. **Unhide** an

element.

2. After **1 second**, **unhide** a element.
3. Change

text to “Hello”.

4. Change text to “**World**”.

```
// illustrative snippet (not full code)
document.querySelector('h1').style.display = 'block';
setTimeout(() => {
  document.querySelector('p').style.display = 'block';
  document.querySelector('h1').textContent = 'Hello';
  document.querySelector('p').textContent = 'World';
}, 1000);
```

JavaScript vs. Java

Core Differences

Aspect	JavaScript	Java
Execution model	Interpreted – runs line-by-line in the browser.	Compiled – source → bytecode → JVM.
Typical use	Front-end web interactivity; now also back-end (Node.js).	Stand-alone applications, Android apps, server-side.
Performance	Historically slower, now highly optimized.	Generally faster due to compilation.
Naming origin	Added “Java” to ride the 1990s Java hype.	Independent language, unrelated to JS.

- The name **JavaScript** was chosen for marketing; Java was “as hot as blockchain is today.”

- The two languages share **very little** beyond superficial syntax; as one professor quipped, they're similar to **cars vs. carpet**.
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JavaScript Popularity

- **Universal browser support**: All major browsers execute JavaScript natively.
- **RedMonk ranking (Jan 2018)** – JavaScript was the **most popular** language, with Java close behind.

Rank (Jan 2018)	Language
1	JavaScript
2	Java



Practical Effects of Disabling JavaScript

Observed Changes (Chrome → Settings → “JavaScript → Block”)

- **Twitter**: Mobile-version loads, but:
 - No dynamic character counter in the compose box.
 - Layout differs drastically from the standard site.
 - **The New York Times**: Ads (which rely on JS) fail to load → **ad-free browsing**.
 - **YouTube / Netflix**: Pages **do not load** at all; core functionality is lost.
 - Some users intentionally disable JavaScript to avoid ads, but they encounter **major usability issues** on most modern sites.
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Resources

- **Bits and Bytes** cartoon (1980s Canadian TV) – explains interpreted vs. compiled languages.
- Link to the **RedMonk** language popularity chart (Jan 2018).

End of segment.